# Huawei HCCDA-AI

# 5-Phase • 100 MCQ Exam Preparation

# THIRD EDITION

100 Fresh Questions — No Mixing with Editions 1 & 2

**Edition rule:** This question bank is newly written for the 3rd Edition. The question wording and tested angles have been changed to avoid repeating the previous two editions.

**Structure:** AI (20) • ML (20) • DL (20) • CV (20) • Huawei Cloud + DevOps/MLOps (20).

**Format:** Question → 4 Options → Correct Answer → Why Correct → Why Every Other Option Is Incorrect.

# Phase 1 — Artificial Intelligence (AI)

20 new questions in this phase.

### Question 1: Which statement best describes the scope of AI?

**A.** It includes methods for building systems that perceive, reason, learn, or act intelligently

**B.** It is limited to database backups

**C.** It is only a hardware manufacturing field

**D.** It only means writing Python loops

**Correct Answer: A. It includes methods for building systems that perceive, reason, learn, or act intelligently**

**Why this is correct:** It includes methods for building systems that perceive, reason, learn, or act intelligently directly matches the concept being tested.

**Why B is incorrect:** It is limited to database backups does not correctly answer the question.

**Why C is incorrect:** It is only a hardware manufacturing field does not correctly answer the question.

**Why D is incorrect:** It only means writing Python loops does not correctly answer the question.

### Question 2: What is meant by Narrow AI?

**A.** An AI system designed for a limited domain or task

**B.** A system with unlimited human-level intelligence

**C.** A cloud storage service

**D.** A type of network cable

**Correct Answer: A. An AI system designed for a limited domain or task**

**Why this is correct:** An AI system designed for a limited domain or task directly matches the concept being tested.

**Why B is incorrect:** A system with unlimited human-level intelligence does not correctly answer the question.

**Why C is incorrect:** A cloud storage service does not correctly answer the question.

**Why D is incorrect:** A type of network cable does not correctly answer the question.

### Question 3: What is a major characteristic of General AI?

**A.** It refers to broad, general-purpose intelligence rather than one narrowly defined task

**B.** It only performs OCR

**C.** It only classifies images

**D.** It is another name for object storage

**Correct Answer: A. It refers to broad, general-purpose intelligence rather than one narrowly defined task**

**Why this is correct:** It refers to broad, general-purpose intelligence rather than one narrowly defined task directly matches the concept being tested.

**Why B is incorrect:** It only performs OCR does not correctly answer the question.

**Why C is incorrect:** It only classifies images does not correctly answer the question.

**Why D is incorrect:** It is another name for object storage does not correctly answer the question.

## Question 4: Why is the historical evolution of AI important to study?

**A.** It helps explain how approaches and capabilities have developed over time

**B.** It determines a computer's screen size

**C.** It replaces practical training

**D.** It changes API authentication

**Correct Answer: A. It helps explain how approaches and capabilities have developed over time**

**Why this is correct:** It helps explain how approaches and capabilities have developed over time directly matches the concept being tested.

**Why B is incorrect:** It determines a computer's screen size does not correctly answer the question.

**Why C is incorrect:** It replaces practical training does not correctly answer the question.

**Why D is incorrect:** It changes API authentication does not correctly answer the question.

## Question 5: Which approach typically represents knowledge using explicit rules?

**A.** Symbolic AI

**B.** Image augmentation

**C.** Cloud storage

**D.** Batch inference

**Correct Answer: A. Symbolic AI**

**Why this is correct:** Symbolic AI directly matches the concept being tested.

**Why B is incorrect:** Image augmentation does not correctly answer the question.

**Why C is incorrect:** Cloud storage does not correctly answer the question.

**Why D is incorrect:** Batch inference does not correctly answer the question.

## Question 6: Which statement best describes learning-based AI?

**A.** A model can infer useful patterns from examples or data

**B.** Every decision must be manually hard-coded

**C.** It cannot use statistical methods

**D.** It only operates on hardware sensors

**Correct Answer: A. A model can infer useful patterns from examples or data**

**Why this is correct:** A model can infer useful patterns from examples or data directly matches the concept being tested.

**Why B is incorrect:** Every decision must be manually hard-coded does not correctly answer the question.

**Why C is incorrect:** It cannot use statistical methods does not correctly answer the question.

**Why D is incorrect:** It only operates on hardware sensors does not correctly answer the question.

## Question 7: Why is data quality important to AI adoption?

**A.** Poor or insufficient data can limit the usefulness of AI systems

**B.** Data quality affects only monitor brightness

**C.** AI systems never use data

**D.** Data quality changes Python indentation

**Correct Answer: A. Poor or insufficient data can limit the usefulness of AI systems**

**Why this is correct:** Poor or insufficient data can limit the usefulness of AI systems directly matches the concept being tested.

**Why B is incorrect:** Data quality affects only monitor brightness does not correctly answer the question.

**Why C is incorrect:** AI systems never use data does not correctly answer the question.

**Why D is incorrect:** Data quality changes Python indentation does not correctly answer the question.

## Question 8: Which challenge can make enterprise AI adoption difficult even when a model works technically?

**A.** Integration with existing systems and workflows

**B.** Choosing a keyboard color

**C.** Changing a PDF font

**D.** Increasing screen size

**Correct Answer: A. Integration with existing systems and workflows**

**Why this is correct:** Integration with existing systems and workflows directly matches the concept being tested.

**Why B is incorrect:** Choosing a keyboard color does not correctly answer the question.

**Why C is incorrect:** Changing a PDF font does not correctly answer the question.

**Why D is incorrect:** Increasing screen size does not correctly answer the question.

## Question 9: Why is an AI industry ecosystem relevant to AI learners?

**A.** It connects technologies, providers, applications, and career opportunities

**B.** It determines the number of pixels in a camera

**C.** It replaces software testing

**D.** It is only a hardware inventory

**Correct Answer: A. It connects technologies, providers, applications, and career opportunities**

**Why this is correct:** It connects technologies, providers, applications, and career opportunities directly matches the concept being tested.

**Why B is incorrect:** It determines the number of pixels in a camera does not correctly answer the question.

**Why C is incorrect:** It replaces software testing does not correctly answer the question.

**Why D is incorrect:** It is only a hardware inventory does not correctly answer the question.

## Question 10: Which capability is associated with Huawei Cloud EI in the supplied course?

**A.** Speech-related AI services

**B.** Operating-system installation

**C.** Printer management

**D.** Spreadsheet macros

**Correct Answer: A. Speech-related AI services**

**Why this is correct:** Speech-related AI services directly matches the concept being tested.

**Why B is incorrect:** Operating-system installation does not correctly answer the question.

**Why C is incorrect:** Printer management does not correctly answer the question.

**Why D is incorrect:** Spreadsheet macros does not correctly answer the question.

## Question 11: What is the purpose of a use-case-to-capability mapping exercise?

**A.** To connect a business requirement with an appropriate AI technology or service

**B.** To remove the business requirement

**C.** To select a classroom monitor

**D.** To avoid architecture planning

**Correct Answer: A. To connect a business requirement with an appropriate AI technology or service**

**Why this is correct:** To connect a business requirement with an appropriate AI technology or service directly matches the concept being tested.

**Why B is incorrect:** To remove the business requirement does not correctly answer the question.

**Why C is incorrect:** To select a classroom monitor does not correctly answer the question.

**Why D is incorrect:** To avoid architecture planning does not correctly answer the question.

## Question 12: What should a well-defined AI system input specification describe?

**A.** The information or data the system expects to receive

**B.** Only the developer's name

**C.** Only the cloud provider

**D.** Only the final certificate

**Correct Answer: A. The information or data the system expects to receive**

**Why this is correct:** The information or data the system expects to receive directly matches the concept being tested.

**Why B is incorrect:** Only the developer's name does not correctly answer the question.

**Why C is incorrect:** Only the cloud provider does not correctly answer the question.

**Why D is incorrect:** Only the final certificate does not correctly answer the question.

## Question 13: What should an AI output specification describe?

**A.** The result the system is expected to produce

**B.** The office location

**C.** The keyboard layout

**D.** The training room size

**Correct Answer: A. The result the system is expected to produce**

**Why this is correct:** The result the system is expected to produce directly matches the concept being tested.

**Why B is incorrect:** The office location does not correctly answer the question.

**Why C is incorrect:** The keyboard layout does not correctly answer the question.

**Why D is incorrect:** The training room size does not correctly answer the question.

## Question 14: Why are non-functional requirements important in AI systems?

**A.** They address qualities such as performance, scalability, and related operational constraints

**B.** They define only image labels

**C.** They replace functional requirements

**D.** They are only about hardware color

**Correct Answer: A. They address qualities such as performance, scalability, and related operational constraints**

**Why this is correct:** They address qualities such as performance, scalability, and related operational constraints directly matches the concept being tested.

**Why B is incorrect:** They define only image labels does not correctly answer the question.

**Why C is incorrect:** They replace functional requirements does not correctly answer the question.

**Why D is incorrect:** They are only about hardware color does not correctly answer the question.

## Question 15: Why should ethical considerations be considered during AI design?

**A.** AI systems can affect people and decisions, so responsible use and risks matter

**B.** Ethics changes CPU frequency

**C.** Ethics removes the need for testing

**D.** Ethics is only a file format

**Correct Answer: A. AI systems can affect people and decisions, so responsible use and risks matter**

**Why this is correct:** AI systems can affect people and decisions, so responsible use and risks matter directly matches the concept being tested.

**Why B is incorrect:** Ethics changes CPU frequency does not correctly answer the question.

**Why C is incorrect:** Ethics removes the need for testing does not correctly answer the question.

**Why D is incorrect:** Ethics is only a file format does not correctly answer the question.

## Question 16: What is a stakeholder in an AI project?

**A.** A person or organization with an interest in or affected by the system

**B.** A Python data type

**C.** A camera pixel

**D.** A cloud storage object

**Correct Answer: A. A person or organization with an interest in or affected by the system**

**Why this is correct:** A person or organization with an interest in or affected by the system directly matches the concept being tested.

**Why B is incorrect:** A Python data type does not correctly answer the question.

**Why C is incorrect:** A camera pixel does not correctly answer the question.

**Why D is incorrect:** A cloud storage object does not correctly answer the question.

## Question 17: What is requirement elicitation?

**A.** The process of discovering and clarifying stakeholder needs and system requirements

**B.** The process of compressing images

**C.** The process of training a CNN only

**D.** The process of creating a monitor

**Correct Answer: A. The process of discovering and clarifying stakeholder needs and system requirements**

**Why this is correct:** The process of discovering and clarifying stakeholder needs and system requirements directly matches the concept being tested.

**Why B is incorrect:** The process of compressing images does not correctly answer the question.

**Why C is incorrect:** The process of training a CNN only does not correctly answer the question.

**Why D is incorrect:** The process of creating a monitor does not correctly answer the question.

## Question 18: Why should the scope of an AI project be controlled?

**A.** A clear scope helps prevent uncontrolled expansion and keeps objectives manageable

**B.** Scope determines keyboard speed

**C.** Scope eliminates the need for requirements

**D.** Scope guarantees model accuracy

**Correct Answer: A. A clear scope helps prevent uncontrolled expansion and keeps objectives manageable**

**Why this is correct:** A clear scope helps prevent uncontrolled expansion and keeps objectives manageable directly matches the concept being tested.

**Why B is incorrect:** Scope determines keyboard speed does not correctly answer the question.

**Why C is incorrect:** Scope eliminates the need for requirements does not correctly answer the question.

**Why D is incorrect:** Scope guarantees model accuracy does not correctly answer the question.

## Question 19: What is one purpose of an architecture decision between edge and cloud?

**A.** To choose where computation should occur based on application requirements

**B.** To select a PDF template

**C.** To choose a programming font

**D.** To rename a dataset

**Correct Answer: A. To choose where computation should occur based on application requirements**

**Why this is correct:** To choose where computation should occur based on application requirements directly matches the concept being tested.

**Why B is incorrect:** To select a PDF template does not correctly answer the question.

**Why C is incorrect:** To choose a programming font does not correctly answer the question.

**Why D is incorrect:** To rename a dataset does not correctly answer the question.

## Question 20: Why is exam-domain weightage useful during certification preparation?

**A.** It helps learners prioritize revision according to the exam structure

**B.** It guarantees the exact questions

**C.** It replaces understanding

**D.** It changes the certification rules

**Correct Answer: A. It helps learners prioritize revision according to the exam structure**

**Why this is correct:** It helps learners prioritize revision according to the exam structure directly matches the concept being tested.

**Why B is incorrect:** It guarantees the exact questions does not correctly answer the question.

**Why C is incorrect:** It replaces understanding does not correctly answer the question.

**Why D is incorrect:** It changes the certification rules does not correctly answer the question.

# Phase 2 — Machine Learning (ML)

20 new questions in this phase.

### Question 1: What is an ML feature?

- **A.** An input variable or representation used by a model to make a prediction
- **B.** A cloud password
- **C.** A monitor setting
- **D.** A PDF page number

**Correct Answer: A. An input variable or representation used by a model to make a prediction**

**Why this is correct:** An input variable or representation used by a model to make a prediction directly matches the concept being tested.

**Why B is incorrect:** A cloud password does not correctly answer the question.

**Why C is incorrect:** A monitor setting does not correctly answer the question.

**Why D is incorrect:** A PDF page number does not correctly answer the question.

### Question 2: What is a target variable?

- **A.** The value or label a supervised model is trained to predict
- **B.** The name of a cloud region
- **C.** A camera lens
- **D.** A source-code comment

**Correct Answer: A. The value or label a supervised model is trained to predict**

**Why this is correct:** The value or label a supervised model is trained to predict directly matches the concept being tested.

**Why B is incorrect:** The name of a cloud region does not correctly answer the question.

**Why C is incorrect:** A camera lens does not correctly answer the question.

**Why D is incorrect:** A source-code comment does not correctly answer the question.

### Question 3: What is data preprocessing?

- **A.** Cleaning and transforming raw data before modeling
- **B.** Deleting the test set
- **C.** Changing hardware
- **D.** Creating a cloud invoice

**Correct Answer: A. Cleaning and transforming raw data before modeling**

**Why this is correct:** Cleaning and transforming raw data before modeling directly matches the concept being tested.

**Why B is incorrect:** Deleting the test set does not correctly answer the question.

**Why C is incorrect:** Changing hardware does not correctly answer the question.

**Why D is incorrect:** Creating a cloud invoice does not correctly answer the question.

## Question 4: Why can categorical data require special handling?

**A.** Many ML algorithms need categorical values represented in a suitable numerical form

**B.** Categorical values are always invalid

**C.** They only affect monitors

**D.** They cannot be stored in Pandas

**Correct Answer: A. Many ML algorithms need categorical values represented in a suitable numerical form**

**Why this is correct:** Many ML algorithms need categorical values represented in a suitable numerical form directly matches the concept being tested.

**Why B is incorrect:** Categorical values are always invalid does not correctly answer the question.

**Why C is incorrect:** They only affect monitors does not correctly answer the question.

**Why D is incorrect:** They cannot be stored in Pandas does not correctly answer the question.

## Question 5: Why is scaling sometimes applied to numerical features?

**A.** It can put features on comparable numerical scales for algorithms sensitive to magnitude

**B.** It always increases the number of classes

**C.** It replaces validation

**D.** It creates labels

**Correct Answer: A. It can put features on comparable numerical scales for algorithms sensitive to magnitude**

**Why this is correct:** It can put features on comparable numerical scales for algorithms sensitive to magnitude directly matches the concept being tested.

**Why B is incorrect:** It always increases the number of classes does not correctly answer the question.

**Why C is incorrect:** It replaces validation does not correctly answer the question.

**Why D is incorrect:** It creates labels does not correctly answer the question.

## Question 6: What does a training algorithm try to minimize or optimize?

**A.** An objective such as a loss function or another defined performance criterion

**B.** Monitor resolution

**C.** File size

**D.** Keyboard latency

**Correct Answer: A. An objective such as a loss function or another defined performance criterion**

**Why this is correct:** An objective such as a loss function or another defined performance criterion directly matches the concept being tested.

**Why B is incorrect:** Monitor resolution does not correctly answer the question.

**Why C is incorrect:** File size does not correctly answer the question.

**Why D is incorrect:** Keyboard latency does not correctly answer the question.

## Question 7: What is a model parameter?

**A.** A value learned from training data by the learning algorithm

**B.** A value typed only into a report

**C.** A cloud billing code

**D.** A filename

**Correct Answer: A. A value learned from training data by the learning algorithm**

**Why this is correct:** A value learned from training data by the learning algorithm directly matches the concept being tested.

**Why B is incorrect:** A value typed only into a report does not correctly answer the question.

**Why C is incorrect:** A cloud billing code does not correctly answer the question.

**Why D is incorrect:** A filename does not correctly answer the question.

## Question 8: What is a hyperparameter search?

**A.** Trying different training configurations to identify settings that work well

**B.** Searching filenames

**C.** Searching cloud regions

**D.** Searching image folders only

**Correct Answer: A. Trying different training configurations to identify settings that work well**

**Why this is correct:** Trying different training configurations to identify settings that work well directly matches the concept being tested.

**Why B is incorrect:** Searching filenames does not correctly answer the question.

**Why C is incorrect:** Searching cloud regions does not correctly answer the question.

**Why D is incorrect:** Searching image folders only does not correctly answer the question.

## Question 9: Why should hyperparameters generally be selected using development data rather than the final test set?

**A.** Using the test set repeatedly can bias the final performance estimate

**B.** The test set cannot contain numbers

**C.** Hyperparameters are hardware settings

**D.** The test set is always deleted

**Correct Answer: A. Using the test set repeatedly can bias the final performance estimate**

**Why this is correct:** Using the test set repeatedly can bias the final performance estimate directly matches the concept being tested.

**Why B is incorrect:** The test set cannot contain numbers does not correctly answer the question.

**Why C is incorrect:** Hyperparameters are hardware settings does not correctly answer the question.

**Why D is incorrect:** The test set is always deleted does not correctly answer the question.

## Question 10: What does cross-validation provide?

**A.** Repeated training/validation splits that can provide a more robust estimate during model selection

**B.** A replacement for all test data

**C.** A cloud API

**D.** An image format

**Correct Answer: A. Repeated training/validation splits that can provide a more robust estimate during model selection**

**Why this is correct:** Repeated training/validation splits that can provide a more robust estimate during model selection directly matches the concept being tested.

**Why B is incorrect:** A replacement for all test data does not correctly answer the question.

**Why C is incorrect:** A cloud API does not correctly answer the question.

**Why D is incorrect:** An image format does not correctly answer the question.

## Question 11: What is a baseline model?

**A.** A simple reference model used as a point of comparison

**B.** The final guaranteed-best model

**C.** A cloud server

**D.** A camera driver

**Correct Answer: A. A simple reference model used as a point of comparison**

**Why this is correct:** A simple reference model used as a point of comparison directly matches the concept being tested.

**Why B is incorrect:** The final guaranteed-best model does not correctly answer the question.

**Why C is incorrect:** A cloud server does not correctly answer the question.

**Why D is incorrect:** A camera driver does not correctly answer the question.

## Question 12: Why establish a baseline before complex optimization?

**A.** It provides a reference for judging whether later changes actually improve results

**B.** It prevents experimentation

**C.** It removes evaluation

**D.** It guarantees high accuracy

**Correct Answer: A. It provides a reference for judging whether later changes actually improve results**

**Why this is correct:** It provides a reference for judging whether later changes actually improve results directly matches the concept being tested.

**Why B is incorrect:** It prevents experimentation does not correctly answer the question.

**Why C is incorrect:** It removes evaluation does not correctly answer the question.

**Why D is incorrect:** It guarantees high accuracy does not correctly answer the question.

## Question 13: What is model latency?

**A.** The time required to produce a prediction or response

**B.** The number of labels

**C.** The image width

**D.** The amount of cloud storage

**Correct Answer: A. The time required to produce a prediction or response**

**Why this is correct:** The time required to produce a prediction or response directly matches the concept being tested.

**Why B is incorrect:** The number of labels does not correctly answer the question.

**Why C is incorrect:** The image width does not correctly answer the question.

**Why D is incorrect:** The amount of cloud storage does not correctly answer the question.

## Question 14: What is throughput?

**A.** The amount of data or requests a system can process in a given period

**B.** The number of classes in a dataset

**C.** The model's accuracy only

**D.** The number of Python files

**Correct Answer: A. The amount of data or requests a system can process in a given period**

**Why this is correct:** The amount of data or requests a system can process in a given period directly matches the concept being tested.

**Why B is incorrect:** The number of classes in a dataset does not correctly answer the question.

**Why C is incorrect:** The model's accuracy only does not correctly answer the question.

**Why D is incorrect:** The number of Python files does not correctly answer the question.

## Question 15: Why can precision and recall need to be considered together?

**A.** They emphasize different types of classification errors and trade-offs

**B.** They measure only CPU speed

**C.** They are both cloud storage metrics

**D.** They always have identical values

**Correct Answer: A. They emphasize different types of classification errors and trade-offs**

**Why this is correct:** They emphasize different types of classification errors and trade-offs directly matches the concept being tested.

**Why B is incorrect:** They measure only CPU speed does not correctly answer the question.

**Why C is incorrect:** They are both cloud storage metrics does not correctly answer the question.

**Why D is incorrect:** They always have identical values does not correctly answer the question.

## Question 16: What is an error analysis report intended to show?

**A.** Patterns and causes behind incorrect predictions

**B.** Only the model filename

**C.** Only the cloud region

**D.** Only the training date

**Correct Answer: A. Patterns and causes behind incorrect predictions**

**Why this is correct:** Patterns and causes behind incorrect predictions directly matches the concept being tested.

**Why B is incorrect:** Only the model filename does not correctly answer the question.

**Why C is incorrect:** Only the cloud region does not correctly answer the question.

**Why D is incorrect:** Only the training date does not correctly answer the question.

## Question 17: Why might targeted samples be added during retraining?

**A.** They can address specific failure cases discovered during evaluation

**B.** They always reduce the dataset quality

**C.** They remove the need for labels

**D.** They disable testing

**Correct Answer: A. They can address specific failure cases discovered during evaluation**

**Why this is correct:** They can address specific failure cases discovered during evaluation directly matches the concept being tested.

**Why B is incorrect:** They always reduce the dataset quality does not correctly answer the question.

**Why C is incorrect:** They remove the need for labels does not correctly answer the question.

**Why D is incorrect:** They disable testing does not correctly answer the question.

## Question 18: What is model reproducibility concerned with?

**A.** Being able to repeat an experiment under controlled versions and settings

**B.** Making images larger

**C.** Changing API URLs

**D.** Replacing the training data with random data

**Correct Answer: A. Being able to repeat an experiment under controlled versions and settings**

**Why this is correct:** Being able to repeat an experiment under controlled versions and settings directly matches the concept being tested.

**Why B is incorrect:** Making images larger does not correctly answer the question.

**Why C is incorrect:** Changing API URLs does not correctly answer the question.

**Why D is incorrect:** Replacing the training data with random data does not correctly answer the question.

## Question 19: Why is traceability valuable in an ML workflow?

**A.** It helps link results to the data, code, configuration, and model version that produced them

**B.** It eliminates logging

**C.** It prevents versioning

**D.** It only changes UI design

**Correct Answer: A. It helps link results to the data, code, configuration, and model version that produced them**

**Why this is correct:** It helps link results to the data, code, configuration, and model version that produced them directly matches the concept being tested.

**Why B is incorrect:** It eliminates logging does not correctly answer the question.

**Why C is incorrect:** It prevents versioning does not correctly answer the question.

**Why D is incorrect:** It only changes UI design does not correctly answer the question.

## Question 20: What is a prediction threshold in a binary classifier?

**A.** A cutoff used to convert a model score or probability into a class decision

**B.** A cloud storage limit

**C.** A camera resolution setting

**D.** A training dataset filename

**Correct Answer: A. A cutoff used to convert a model score or probability into a class decision**

**Why this is correct:** A cutoff used to convert a model score or probability into a class decision directly matches the concept being tested.

**Why B is incorrect:** A cloud storage limit does not correctly answer the question.

**Why C is incorrect:** A camera resolution setting does not correctly answer the question.

**Why D is incorrect:** A training dataset filename does not correctly answer the question.

# Phase 3 — Deep Learning (DL)

20 new questions in this phase.

### Question 1: What happens during one forward pass of a neural network?

**A.** Inputs are transformed through the network to calculate an output

**B.** Weights are permanently deleted

**C.** The dataset is renamed

**D.** OBS is formatted

**Correct Answer: A. Inputs are transformed through the network to calculate an output**

**Why this is correct:** Inputs are transformed through the network to calculate an output directly matches the concept being tested.

**Why B is incorrect:** Weights are permanently deleted does not correctly answer the question.

**Why C is incorrect:** The dataset is renamed does not correctly answer the question.

**Why D is incorrect:** OBS is formatted does not correctly answer the question.

### Question 2: What is a gradient in neural-network training?

**A.** A measure indicating how changing parameters affects the objective

**B.** A cloud storage object

**C.** An image filename

**D.** A camera setting

**Correct Answer: A. A measure indicating how changing parameters affects the objective**

**Why this is correct:** A measure indicating how changing parameters affects the objective directly matches the concept being tested.

**Why B is incorrect:** A cloud storage object does not correctly answer the question.

**Why C is incorrect:** An image filename does not correctly answer the question.

**Why D is incorrect:** A camera setting does not correctly answer the question.

### Question 3: Why are gradients important?

**A.** They provide information used by optimizers to update learnable parameters

**B.** They create image files

**C.** They replace labels

**D.** They configure networks

**Correct Answer: A. They provide information used by optimizers to update learnable parameters**

**Why this is correct:** They provide information used by optimizers to update learnable parameters directly matches the concept being tested.

**Why B is incorrect:** They create image files does not correctly answer the question.

**Why C is incorrect:** They replace labels does not correctly answer the question.

**Why D is incorrect:** They configure networks does not correctly answer the question.

## Question 4: What is the learning rate?

**A.** A training hyperparameter controlling the size of parameter updates

**B.** The number of images in a dataset

**C.** The API response time

**D.** The number of cloud regions

**Correct Answer: A. A training hyperparameter controlling the size of parameter updates**

**Why this is correct:** A training hyperparameter controlling the size of parameter updates directly matches the concept being tested.

**Why B is incorrect:** The number of images in a dataset does not correctly answer the question.

**Why C is incorrect:** The API response time does not correctly answer the question.

**Why D is incorrect:** The number of cloud regions does not correctly answer the question.

## Question 5: What can happen if a learning rate is too large?

**A.** Training can become unstable or fail to converge effectively

**B.** The image automatically becomes sharper

**C.** The dataset becomes larger

**D.** OCR becomes perfect

**Correct Answer: A. Training can become unstable or fail to converge effectively**

**Why this is correct:** Training can become unstable or fail to converge effectively directly matches the concept being tested.

**Why B is incorrect:** The image automatically becomes sharper does not correctly answer the question.

**Why C is incorrect:** The dataset becomes larger does not correctly answer the question.

**Why D is incorrect:** OCR becomes perfect does not correctly answer the question.

## Question 6: What can happen if a learning rate is too small?

**A.** Training may progress very slowly

**B.** The model always overfits immediately

**C.** The cloud account expires

**D.** The camera changes resolution

**Correct Answer: A. Training may progress very slowly**

**Why this is correct:** Training may progress very slowly directly matches the concept being tested.

**Why B is incorrect:** The model always overfits immediately does not correctly answer the question.

**Why C is incorrect:** The cloud account expires does not correctly answer the question.

**Why D is incorrect:** The camera changes resolution does not correctly answer the question.

## Question 7: What is dropout?

**A.** A regularization technique that randomly omits units during training

**B.** A cloud storage service

**C.** A computer vision detector

**D.** An API authentication method

**Correct Answer: A. A regularization technique that randomly omits units during training**

**Why this is correct:** A regularization technique that randomly omits units during training directly matches the concept being tested.

**Why B is incorrect:** A cloud storage service does not correctly answer the question.

**Why C is incorrect:** A computer vision detector does not correctly answer the question.

**Why D is incorrect:** An API authentication method does not correctly answer the question.

## Question 8: Why can dropout help generalization?

**A.** It can reduce reliance on specific neuron combinations during training

**B.** It guarantees zero loss

**C.** It increases image resolution

**D.** It removes validation

**Correct Answer: A. It can reduce reliance on specific neuron combinations during training**

**Why this is correct:** It can reduce reliance on specific neuron combinations during training directly matches the concept being tested.

**Why B is incorrect:** It guarantees zero loss does not correctly answer the question.

**Why C is incorrect:** It increases image resolution does not correctly answer the question.

**Why D is incorrect:** It removes validation does not correctly answer the question.

## Question 9: What is batch normalization intended to do?

**A.** Normalize intermediate activations in a neural network in a training-aware way

**B.** Store datasets in OBS

**C.** Perform OCR

**D.** Create API endpoints

**Correct Answer: A. Normalize intermediate activations in a neural network in a training-aware way**

**Why this is correct:** Normalize intermediate activations in a neural network in a training-aware way directly matches the concept being tested.

**Why B is incorrect:** Store datasets in OBS does not correctly answer the question.

**Why C is incorrect:** Perform OCR does not correctly answer the question.

**Why D is incorrect:** Create API endpoints does not correctly answer the question.

## Question 10: Why can residual connections help very deep networks?

**A.** They provide shortcut paths that can make optimization and gradient flow easier

**B.** They remove all parameters

**C.** They convert images to text

**D.** They replace datasets

**Correct Answer: A. They provide shortcut paths that can make optimization and gradient flow easier**

**Why this is correct:** They provide shortcut paths that can make optimization and gradient flow easier directly matches the concept being tested.

**Why B is incorrect:** They remove all parameters does not correctly answer the question.

**Why C is incorrect:** They convert images to text does not correctly answer the question.

**Why D is incorrect:** They replace datasets does not correctly answer the question.

## Question 11: What is a convolution kernel size?

**A.** The spatial dimensions of the learnable filter applied to an input

**B.** The number of cloud accounts

**C.** The dataset file size

**D.** The number of classes

**Correct Answer: A. The spatial dimensions of the learnable filter applied to an input**

**Why this is correct:** The spatial dimensions of the learnable filter applied to an input directly matches the concept being tested.

**Why B is incorrect:** The number of cloud accounts does not correctly answer the question.

**Why C is incorrect:** The dataset file size does not correctly answer the question.

**Why D is incorrect:** The number of classes does not correctly answer the question.

## Question 12: What does a CNN receptive field describe?

**A.** The region of the input that can influence a particular feature response

**B.** The model's cloud bill

**C.** The API endpoint name

**D.** The number of Python functions

**Correct Answer: A. The region of the input that can influence a particular feature response**

**Why this is correct:** The region of the input that can influence a particular feature response directly matches the concept being tested.

**Why B is incorrect:** The model's cloud bill does not correctly answer the question.

**Why C is incorrect:** The API endpoint name does not correctly answer the question.

**Why D is incorrect:** The number of Python functions does not correctly answer the question.

## Question 13: Why can deeper CNN layers represent more abstract features?

**A.** They combine information from earlier learned representations

**B.** They only see raw filenames

**C.** They cannot use convolution

**D.** They remove all spatial information immediately

**Correct Answer: A. They combine information from earlier learned representations**

**Why this is correct:** They combine information from earlier learned representations directly matches the concept being tested.

**Why B is incorrect:** They only see raw filenames does not correctly answer the question.

**Why C is incorrect:** They cannot use convolution does not correctly answer the question.

**Why D is incorrect:** They remove all spatial information immediately does not correctly answer the question.

## Question 14: What is an RNN hidden state intended to represent?

**A.** Information carried forward from previous sequence steps

**B.** A cloud bucket

**C.** A camera calibration value

**D.** An image file extension

**Correct Answer: A. Information carried forward from previous sequence steps**

**Why this is correct:** Information carried forward from previous sequence steps directly matches the concept being tested.

**Why B is incorrect:** A cloud bucket does not correctly answer the question.

**Why C is incorrect:** A camera calibration value does not correctly answer the question.

**Why D is incorrect:** An image file extension does not correctly answer the question.

## Question 15: Why are RNNs relevant to sequential data?

**A.** Their recurrent structure can incorporate information from earlier steps

**B.** They only process static images

**C.** They cannot process sequences

**D.** They are storage systems

**Correct Answer: A. Their recurrent structure can incorporate information from earlier steps**

**Why this is correct:** Their recurrent structure can incorporate information from earlier steps directly matches the concept being tested.

**Why B is incorrect:** They only process static images does not correctly answer the question.

**Why C is incorrect:** They cannot process sequences does not correctly answer the question.

**Why D is incorrect:** They are storage systems does not correctly answer the question.

## Question 16: What is an epoch count used for in training?

**A.** It specifies how many complete passes over the training data are performed

**B.** It sets the number of cloud buckets

**C.** It sets the number of classes

**D.** It defines image color

**Correct Answer: A. It specifies how many complete passes over the training data are performed**

**Why this is correct:** It specifies how many complete passes over the training data are performed directly matches the concept being tested.

**Why B is incorrect:** It sets the number of cloud buckets does not correctly answer the question.

**Why C is incorrect:** It sets the number of classes does not correctly answer the question.

**Why D is incorrect:** It defines image color does not correctly answer the question.

## Question 17: Why save checkpoints during deep-learning training?

**A.** To preserve useful model states and enable recovery or comparison

**B.** To change image labels

**C.** To replace evaluation

**D.** To increase network bandwidth

**Correct Answer: A. To preserve useful model states and enable recovery or comparison**

**Why this is correct:** To preserve useful model states and enable recovery or comparison directly matches the concept being tested.

**Why B is incorrect:** To change image labels does not correctly answer the question.

**Why C is incorrect:** To replace evaluation does not correctly answer the question.

**Why D is incorrect:** To increase network bandwidth does not correctly answer the question.

## Question 18: What is a pretrained model's learned representation?

**A.** Features or internal patterns learned from its original training data

**B.** A list of API keys

**C.** A set of cloud invoices

**D.** A camera specification

**Correct Answer: A. Features or internal patterns learned from its original training data**

**Why this is correct:** Features or internal patterns learned from its original training data directly matches the concept being tested.

**Why B is incorrect:** A list of API keys does not correctly answer the question.

**Why C is incorrect:** A set of cloud invoices does not correctly answer the question.

**Why D is incorrect:** A camera specification does not correctly answer the question.

## Question 19: Why evaluate a fine-tuned model on held-out data?

**A.** To determine whether adaptation generalizes beyond the fine-tuning examples

**B.** To change its architecture automatically

**C.** To eliminate overfitting

**D.** To create a new dataset

**Correct Answer: A. To determine whether adaptation generalizes beyond the fine-tuning examples**

**Why this is correct:** To determine whether adaptation generalizes beyond the fine-tuning examples directly matches the concept being tested.

**Why B is incorrect:** To change its architecture automatically does not correctly answer the question.

**Why C is incorrect:** To eliminate overfitting does not correctly answer the question.

**Why D is incorrect:** To create a new dataset does not correctly answer the question.

## Question 20: Why is framework choice relevant in deep-learning development?

**A.** APIs, tooling, deployment support, and developer workflow can differ between frameworks

**B.** Frameworks only change monitor resolution

**C.** All frameworks have identical capabilities

**D.** Framework choice has no effect on development

**Correct Answer: A. APIs, tooling, deployment support, and developer workflow can differ between frameworks**

**Why this is correct:** APIs, tooling, deployment support, and developer workflow can differ between frameworks directly matches the concept being tested.

**Why B is incorrect:** Frameworks only change monitor resolution does not correctly answer the question.

**Why C is incorrect:** All frameworks have identical capabilities does not correctly answer the question.

**Why D is incorrect:** Framework choice has no effect on development does not correctly answer the question.

# Phase 4 — Computer Vision (CV)

20 new questions in this phase.

### Question 1: What does an RGB image normally contain?

**A.** Three color channels representing red, green, and blue components

**B.** Only one text channel

**C.** Only a depth channel

**D.** Only an audio channel

**Correct Answer: A. Three color channels representing red, green, and blue components**

**Why this is correct:** Three color channels representing red, green, and blue components directly matches the concept being tested.

**Why B is incorrect:** Only one text channel does not correctly answer the question.

**Why C is incorrect:** Only a depth channel does not correctly answer the question.

**Why D is incorrect:** Only an audio channel does not correctly answer the question.

### Question 2: Why might BGR appear when working with OpenCV?

**A.** OpenCV commonly represents loaded color images in BGR channel order

**B.** BGR is a cloud API

**C.** BGR is a neural-network optimizer

**D.** BGR is a segmentation metric

**Correct Answer: A. OpenCV commonly represents loaded color images in BGR channel order**

**Why this is correct:** OpenCV commonly represents loaded color images in BGR channel order directly matches the concept being tested.

**Why B is incorrect:** BGR is a cloud API does not correctly answer the question.

**Why C is incorrect:** BGR is a neural-network optimizer does not correctly answer the question.

**Why D is incorrect:** BGR is a segmentation metric does not correctly answer the question.

### Question 3: What is an image histogram?

**A.** A representation of the distribution of pixel intensity or color values

**B.** A cloud deployment pipeline

**C.** A model checkpoint

**D.** A database schema

**Correct Answer: A. A representation of the distribution of pixel intensity or color values**

**Why this is correct:** A representation of the distribution of pixel intensity or color values directly matches the concept being tested.

**Why B is incorrect:** A cloud deployment pipeline does not correctly answer the question.

**Why C is incorrect:** A model checkpoint does not correctly answer the question.

**Why D is incorrect:** A database schema does not correctly answer the question.

## Question 4: Why can histogram information be useful?

**A.** It can help analyze brightness or contrast characteristics of an image

**B.** It automatically detects every object

**C.** It replaces annotations

**D.** It performs OCR without a model

**Correct Answer: A. It can help analyze brightness or contrast characteristics of an image**

**Why this is correct:** It can help analyze brightness or contrast characteristics of an image directly matches the concept being tested.

**Why B is incorrect:** It automatically detects every object does not correctly answer the question.

**Why C is incorrect:** It replaces annotations does not correctly answer the question.

**Why D is incorrect:** It performs OCR without a model does not correctly answer the question.

## Question 5: What is contrast enhancement intended to improve?

**A.** The visual distinction between intensity levels or regions

**B.** The number of cloud instances

**C.** The number of labels

**D.** The API credentials

**Correct Answer: A. The visual distinction between intensity levels or regions**

**Why this is correct:** The visual distinction between intensity levels or regions directly matches the concept being tested.

**Why B is incorrect:** The number of cloud instances does not correctly answer the question.

**Why C is incorrect:** The number of labels does not correctly answer the question.

**Why D is incorrect:** The API credentials does not correctly answer the question.

## Question 6: What is Gaussian smoothing commonly used for?

**A.** Reducing high-frequency noise before certain image-processing operations

**B.** Creating object labels

**C.** Deploying models

**D.** Calling REST APIs

**Correct Answer: A. Reducing high-frequency noise before certain image-processing operations**

**Why this is correct:** Reducing high-frequency noise before certain image-processing operations directly matches the concept being tested.

**Why B is incorrect:** Creating object labels does not correctly answer the question.

**Why C is incorrect:** Deploying models does not correctly answer the question.

**Why D is incorrect:** Calling REST APIs does not correctly answer the question.

## Question 7: Why is edge detection useful?

**A.** Edges can highlight boundaries where image intensity changes significantly

**B.** It directly identifies every object class

**C.** It stores images in OBS

**D.** It replaces segmentation

**Correct Answer: A. Edges can highlight boundaries where image intensity changes significantly**

**Why this is correct:** Edges can highlight boundaries where image intensity changes significantly directly matches the concept being tested.

**Why B is incorrect:** It directly identifies every object class does not correctly answer the question.

**Why C is incorrect:** It stores images in OBS does not correctly answer the question.

**Why D is incorrect:** It replaces segmentation does not correctly answer the question.

## Question 8: What is a contour in image processing?

**A.** A curve or boundary representing the shape of a connected region

**B.** A cloud endpoint

**C.** A neural-network weight

**D.** An API token

**Correct Answer: A. A curve or boundary representing the shape of a connected region**

**Why this is correct:** A curve or boundary representing the shape of a connected region directly matches the concept being tested.

**Why B is incorrect:** A cloud endpoint does not correctly answer the question.

**Why C is incorrect:** A neural-network weight does not correctly answer the question.

**Why D is incorrect:** An API token does not correctly answer the question.

## Question 9: What is connected-component analysis used for?

**A.** Identifying separate connected regions in a binary or labeled image

**B.** Training an RNN

**C.** Creating cloud credentials

**D.** Changing image file extensions

**Correct Answer: A. Identifying separate connected regions in a binary or labeled image**

**Why this is correct:** Identifying separate connected regions in a binary or labeled image directly matches the concept being tested.

**Why B is incorrect:** Training an RNN does not correctly answer the question.

**Why C is incorrect:** Creating cloud credentials does not correctly answer the question.

**Why D is incorrect:** Changing image file extensions does not correctly answer the question.

## Question 10: Why are morphological operations useful after thresholding?

**A.** They can clean or modify binary regions by expanding, shrinking, opening, or closing structures

**B.** They replace all CNN training

**C.** They create cloud storage

**D.** They generate audio

**Correct Answer: A. They can clean or modify binary regions by expanding, shrinking, opening, or closing structures**

**Why this is correct:** They can clean or modify binary regions by expanding, shrinking, opening, or closing structures directly matches the concept being tested.

**Why B is incorrect:** They replace all CNN training does not correctly answer the question.

**Why C is incorrect:** They create cloud storage does not correctly answer the question.

**Why D is incorrect:** They generate audio does not correctly answer the question.

## Question 11: What is a binary image?

**A.** An image whose pixels are represented using two states or classes

**B.** An image with exactly two cameras

**C.** An image with two files

**D.** A cloud object

**Correct Answer: A. An image whose pixels are represented using two states or classes**

**Why this is correct:** An image whose pixels are represented using two states or classes directly matches the concept being tested.

**Why B is incorrect:** An image with exactly two cameras does not correctly answer the question.

**Why C is incorrect:** An image with two files does not correctly answer the question.

**Why D is incorrect:** A cloud object does not correctly answer the question.

## Question 12: What is perspective distortion?

**A.** A change in apparent geometry caused by viewpoint and projection

**B.** A database error

**C.** A model checkpoint

**D.** An OCR credential

**Correct Answer: A. A change in apparent geometry caused by viewpoint and projection**

**Why this is correct:** A change in apparent geometry caused by viewpoint and projection directly matches the concept being tested.

**Why B is incorrect:** A database error does not correctly answer the question.

**Why C is incorrect:** A model checkpoint does not correctly answer the question.

**Why D is incorrect:** An OCR credential does not correctly answer the question.

## Question 13: Why is perspective correction useful for document images?

**A.** It can make a photographed document more geometrically aligned for later processing

**B.** It increases CPU speed

**C.** It changes the document text

**D.** It replaces OCR

**Correct Answer: A. It can make a photographed document more geometrically aligned for later processing**

**Why this is correct:** It can make a photographed document more geometrically aligned for later processing directly matches the concept being tested.

**Why B is incorrect:** It increases CPU speed does not correctly answer the question.

**Why C is incorrect:** It changes the document text does not correctly answer the question.

**Why D is incorrect:** It replaces OCR does not correctly answer the question.

## Question 14: What is face landmark information useful for?

**A.** Representing key facial locations for tasks such as alignment or expression analysis

**B.** Storing cloud passwords

**C.** Detecting network routes

**D.** Measuring disk capacity

**Correct Answer: A. Representing key facial locations for tasks such as alignment or expression analysis**

**Why this is correct:** Representing key facial locations for tasks such as alignment or expression analysis directly matches the concept being tested.

**Why B is incorrect:** Storing cloud passwords does not correctly answer the question.

**Why C is incorrect:** Detecting network routes does not correctly answer the question.

**Why D is incorrect:** Measuring disk capacity does not correctly answer the question.

## Question 15: What is the difference between face detection and face verification?

**A.** Detection locates faces, while verification checks whether two face representations match a claimed identity

**B.** They are identical tasks

**C.** Verification only finds image edges

**D.** Detection requires no image

**Correct Answer: A. Detection locates faces, while verification checks whether two face representations match a claimed identity**

**Why this is correct:** Detection locates faces, while verification checks whether two face representations match a claimed identity directly matches the concept being tested.

**Why B is incorrect:** They are identical tasks does not correctly answer the question.

**Why C is incorrect:** Verification only finds image edges does not correctly answer the question.

**Why D is incorrect:** Detection requires no image does not correctly answer the question.

## Question 16: What is a feature descriptor in classical computer vision?

**A.** A numerical representation of local visual characteristics used for matching or recognition

**B.** A cloud storage file

**C.** A Python package manager

**D.** A deployment endpoint

**Correct Answer: A. A numerical representation of local visual characteristics used for matching or recognition**

**Why this is correct:** A numerical representation of local visual characteristics used for matching or recognition directly matches the concept being tested.

**Why B is incorrect:** A cloud storage file does not correctly answer the question.

**Why C is incorrect:** A Python package manager does not correctly answer the question.

**Why D is incorrect:** A deployment endpoint does not correctly answer the question.

## Question 17: Why are local feature methods useful for image matching?

**A.** They provide representations that can be compared between images

**B.** They guarantee perfect matching

**C.** They replace camera calibration

**D.** They only work on audio

**Correct Answer: A. They provide representations that can be compared between images**

**Why this is correct:** They provide representations that can be compared between images directly matches the concept being tested.

**Why B is incorrect:** They guarantee perfect matching does not correctly answer the question.

**Why C is incorrect:** They replace camera calibration does not correctly answer the question.

**Why D is incorrect:** They only work on audio does not correctly answer the question.

## Question 18: What is camera pose?

**A.** The camera's position and orientation relative to a scene

**B.** The camera's brand name

**C.** The image filename

**D.** The cloud API key

**Correct Answer: A. The camera's position and orientation relative to a scene**

**Why this is correct:** The camera's position and orientation relative to a scene directly matches the concept being tested.

**Why B is incorrect:** The camera's brand name does not correctly answer the question.

**Why C is incorrect:** The image filename does not correctly answer the question.

**Why D is incorrect:** The cloud API key does not correctly answer the question.

## Question 19: Why can camera calibration matter in computer vision?

**A.** Accurate camera parameters improve geometric interpretation of image measurements

**B.** It changes image labels automatically

**C.** It replaces object detection

**D.** It increases storage capacity

**Correct Answer: A. Accurate camera parameters improve geometric interpretation of image measurements**

**Why this is correct:** Accurate camera parameters improve geometric interpretation of image measurements directly matches the concept being tested.

**Why B is incorrect:** It changes image labels automatically does not correctly answer the question.

**Why C is incorrect:** It replaces object detection does not correctly answer the question.

**Why D is incorrect:** It increases storage capacity does not correctly answer the question.

## Question 20: What is depth information useful for?

**A.** Estimating spatial relationships and distances within a scene

**B.** Changing text fonts

**C.** Creating API credentials

**D.** Replacing image pixels

**Correct Answer: A. Estimating spatial relationships and distances within a scene**

**Why this is correct:** Estimating spatial relationships and distances within a scene directly matches the concept being tested.

**Why B is incorrect:** Changing text fonts does not correctly answer the question.

**Why C is incorrect:** Creating API credentials does not correctly answer the question.

**Why D is incorrect:** Replacing image pixels does not correctly answer the question.

# Phase 5 — Huawei Cloud + DevOps/MLOps

20 new questions in this phase.

### Question 1: What is an OBS bucket?

**A.** A logical container in object storage for storing objects such as datasets

**B.** A neural-network layer

**C.** A camera frame

**D.** A Python class

**Correct Answer: A. A logical container in object storage for storing objects such as datasets**

**Why this is correct:** A logical container in object storage for storing objects such as datasets directly matches the concept being tested.

**Why B is incorrect:** A neural-network layer does not correctly answer the question.

**Why C is incorrect:** A camera frame does not correctly answer the question.

**Why D is incorrect:** A Python class does not correctly answer the question.

### Question 2: Why might a ModelArts notebook access OBS?

**A.** To read or write datasets and other AI workflow artifacts stored in cloud storage

**B.** To change the monitor resolution

**C.** To train without data

**D.** To replace an API

**Correct Answer: A. To read or write datasets and other AI workflow artifacts stored in cloud storage**

**Why this is correct:** To read or write datasets and other AI workflow artifacts stored in cloud storage directly matches the concept being tested.

**Why B is incorrect:** To change the monitor resolution does not correctly answer the question.

**Why C is incorrect:** To train without data does not correctly answer the question.

**Why D is incorrect:** To replace an API does not correctly answer the question.

### Question 3: What is AutoML experiment comparison useful for?

**A.** Comparing multiple automatically or systematically generated model runs

**B.** Managing classroom attendance

**C.** Changing image colors

**D.** Creating hardware drivers

**Correct Answer: A. Comparing multiple automatically or systematically generated model runs**

**Why this is correct:** Comparing multiple automatically or systematically generated model runs directly matches the concept being tested.

**Why B is incorrect:** Managing classroom attendance does not correctly answer the question.

**Why C is incorrect:** Changing image colors does not correctly answer the question.

**Why D is incorrect:** Creating hardware drivers does not correctly answer the question.

## Question 4: What does model deployment mean?

**A.** Making a trained model available for inference in a target environment

**B.** Deleting the trained model

**C.** Collecting keyboard data

**D.** Renaming an image

**Correct Answer: A. Making a trained model available for inference in a target environment**

**Why this is correct:** Making a trained model available for inference in a target environment directly matches the concept being tested.

**Why B is incorrect:** Deleting the trained model does not correctly answer the question.

**Why C is incorrect:** Collecting keyboard data does not correctly answer the question.

**Why D is incorrect:** Renaming an image does not correctly answer the question.

## Question 5: When is batch inference often appropriate?

**A.** When many inputs can be processed without requiring an immediate individual response

**B.** When every prediction must be instantaneous

**C.** Only when no model exists

**D.** Only for camera calibration

**Correct Answer: A. When many inputs can be processed without requiring an immediate individual response**

**Why this is correct:** When many inputs can be processed without requiring an immediate individual response directly matches the concept being tested.

**Why B is incorrect:** When every prediction must be instantaneous does not correctly answer the question.

**Why C is incorrect:** Only when no model exists does not correctly answer the question.

**Why D is incorrect:** Only for camera calibration does not correctly answer the question.

## Question 6: When is real-time inference often appropriate?

**A.** When an application needs predictions in response to individual requests with low delay

**B.** Only for historical reports

**C.** Only for offline storage

**D.** Only for source-code review

**Correct Answer: A. When an application needs predictions in response to individual requests with low delay**

**Why this is correct:** When an application needs predictions in response to individual requests with low delay directly matches the concept being tested.

**Why B is incorrect:** Only for historical reports does not correctly answer the question.

**Why C is incorrect:** Only for offline storage does not correctly answer the question.

**Why D is incorrect:** Only for source-code review does not correctly answer the question.

## Question 7: What is API versioning intended to support?

**A.** Managing changes to an interface while reducing unexpected impact on clients

**B.** Changing monitor drivers

**C.** Deleting models

**D.** Replacing training labels

**Correct Answer: A. Managing changes to an interface while reducing unexpected impact on clients**

**Why this is correct:** Managing changes to an interface while reducing unexpected impact on clients directly matches the concept being tested.

**Why B is incorrect:** Changing monitor drivers does not correctly answer the question.

**Why C is incorrect:** Deleting models does not correctly answer the question.

**Why D is incorrect:** Replacing training labels does not correctly answer the question.

## Question 8: Why should deployed model APIs validate incoming input?

**A.** Invalid or unexpected input can cause errors, unsafe behavior, or poor predictions

**B.** Validation is unnecessary in production

**C.** It changes model architecture

**D.** It creates new GPUs

**Correct Answer: A. Invalid or unexpected input can cause errors, unsafe behavior, or poor predictions**

**Why this is correct:** Invalid or unexpected input can cause errors, unsafe behavior, or poor predictions directly matches the concept being tested.

**Why B is incorrect:** Validation is unnecessary in production does not correctly answer the question.

**Why C is incorrect:** It changes model architecture does not correctly answer the question.

**Why D is incorrect:** It creates new GPUs does not correctly answer the question.

## Question 9: What is a CI pipeline typically triggered by?

**A.** A code or repository event such as a commit or pull request

**B.** A monitor brightness change

**C.** A camera lens change

**D.** A PDF page count

**Correct Answer: A. A code or repository event such as a commit or pull request**

**Why this is correct:** A code or repository event such as a commit or pull request directly matches the concept being tested.

**Why B is incorrect:** A monitor brightness change does not correctly answer the question.

**Why C is incorrect:** A camera lens change does not correctly answer the question.

**Why D is incorrect:** A PDF page count does not correctly answer the question.

## Question 10: What is continuous delivery concerned with?

**A.** Keeping software changes in a releasable state through automated processes

**B.** Training only once

**C.** Deleting test results

**D.** Managing office hardware

**Correct Answer: A. Keeping software changes in a releasable state through automated processes**

**Why this is correct:** Keeping software changes in a releasable state through automated processes directly matches the concept being tested.

**Why B is incorrect:** Training only once does not correctly answer the question.

**Why C is incorrect:** Deleting test results does not correctly answer the question.

**Why D is incorrect:** Managing office hardware does not correctly answer the question.

## Question 11: Why integrate tests into CI/CD?

**A.** To catch defects earlier and reduce the chance of broken changes being delivered

**B.** To eliminate all human review

**C.** To increase image resolution

**D.** To replace monitoring

**Correct Answer: A. To catch defects earlier and reduce the chance of broken changes being delivered**

**Why this is correct:** To catch defects earlier and reduce the chance of broken changes being delivered directly matches the concept being tested.

**Why B is incorrect:** To eliminate all human review does not correctly answer the question.

**Why C is incorrect:** To increase image resolution does not correctly answer the question.

**Why D is incorrect:** To replace monitoring does not correctly answer the question.

## Question 12: What is an ML pipeline artifact?

**A.** A produced or stored item such as a dataset version, model, metric report, or package

**B.** A keyboard key

**C.** A monitor pixel

**D.** A camera lens

**Correct Answer: A. A produced or stored item such as a dataset version, model, metric report, or package**

**Why this is correct:** A produced or stored item such as a dataset version, model, metric report, or package directly matches the concept being tested.

**Why B is incorrect:** A keyboard key does not correctly answer the question.

**Why C is incorrect:** A monitor pixel does not correctly answer the question.

**Why D is incorrect:** A camera lens does not correctly answer the question.

## Question 13: Why should model versions be tracked?

**A.** To identify which model is deployed and compare it with previous versions

**B.** To change cloud pricing

**C.** To remove all logs

**D.** To avoid testing

**Correct Answer: A. To identify which model is deployed and compare it with previous versions**

**Why this is correct:** To identify which model is deployed and compare it with previous versions directly matches the concept being tested.

**Why B is incorrect:** To change cloud pricing does not correctly answer the question.

**Why C is incorrect:** To remove all logs does not correctly answer the question.

**Why D is incorrect:** To avoid testing does not correctly answer the question.

## Question 14: What is a production rollback?

**A.** Returning a system to a previously known-good deployed version

**B.** Deleting all backups

**C.** Changing a model's class labels

**D.** Creating a new dataset

**Correct Answer: A. Returning a system to a previously known-good deployed version**

**Why this is correct:** Returning a system to a previously known-good deployed version directly matches the concept being tested.

**Why B is incorrect:** Deleting all backups does not correctly answer the question.

**Why C is incorrect:** Changing a model's class labels does not correctly answer the question.

**Why D is incorrect:** Creating a new dataset does not correctly answer the question.

## Question 15: Why can monitoring include both system and model metrics?

**A.** Operational health and model quality can fail independently

**B.** Only CPU metrics matter

**C.** Only accuracy matters

**D.** Neither needs monitoring

**Correct Answer: A. Operational health and model quality can fail independently**

**Why this is correct:** Operational health and model quality can fail independently directly matches the concept being tested.

**Why B is incorrect:** Only CPU metrics matter does not correctly answer the question.

**Why C is incorrect:** Only accuracy matters does not correctly answer the question.

**Why D is incorrect:** Neither needs monitoring does not correctly answer the question.

## Question 16: What is an alert threshold?

**A.** A defined value beyond which a monitoring system should raise an alert

**B.** A model label

**C.** A dataset filename

**D.** A cloud storage object

**Correct Answer: A. A defined value beyond which a monitoring system should raise an alert**

**Why this is correct:** A defined value beyond which a monitoring system should raise an alert directly matches the concept being tested.

**Why B is incorrect:** A model label does not correctly answer the question.

**Why C is incorrect:** A dataset filename does not correctly answer the question.

**Why D is incorrect:** A cloud storage object does not correctly answer the question.

## Question 17: Why is retraining not automatically appropriate for every alert?

**A.** Alerts should be investigated to distinguish true drift or failures from temporary or unrelated issues

**B.** Retraining always harms accuracy

**C.** Alerts never contain useful information

**D.** Retraining cannot use data

**Correct Answer: A. Alerts should be investigated to distinguish true drift or failures from temporary or unrelated issues**

**Why this is correct:** Alerts should be investigated to distinguish true drift or failures from temporary or unrelated issues directly matches the concept being tested.

**Why B is incorrect:** Retraining always harms accuracy does not correctly answer the question.

**Why C is incorrect:** Alerts never contain useful information does not correctly answer the question.

**Why D is incorrect:** Retraining cannot use data does not correctly answer the question.

## Question 18: What does reproducible MLOps require?

**A.** Consistent tracking of code, data, configuration, and model artifacts

**B.** Only a model filename

**C.** Only a cloud password

**D.** Only a screenshot

**Correct Answer: A. Consistent tracking of code, data, configuration, and model artifacts**

**Why this is correct:** Consistent tracking of code, data, configuration, and model artifacts directly matches the concept being tested.

**Why B is incorrect:** Only a model filename does not correctly answer the question.

**Why C is incorrect:** Only a cloud password does not correctly answer the question.

**Why D is incorrect:** Only a screenshot does not correctly answer the question.

## Question 19: Why is traceability important after a model is deployed?

**A.** Teams may need to determine which data, code, configuration, and model produced a result

**B.** It changes the model automatically

**C.** It removes all incidents

**D.** It eliminates monitoring

**Correct Answer: A. Teams may need to determine which data, code, configuration, and model produced a result**

**Why this is correct:** Teams may need to determine which data, code, configuration, and model produced a result directly matches the concept being tested.

**Why B is incorrect:** It changes the model automatically does not correctly answer the question.

**Why C is incorrect:** It removes all incidents does not correctly answer the question.

**Why D is incorrect:** It eliminates monitoring does not correctly answer the question.

## Question 20: What is the purpose of a capstone AI project in the supplied course?

**A.** To integrate learning into a practical solution and demonstrate implementation and results

**B.** To avoid using AI tools

**C.** To replace certification study entirely

**D.** To focus only on hardware

**Correct Answer: A. To integrate learning into a practical solution and demonstrate implementation and results**

**Why this is correct:** To integrate learning into a practical solution and demonstrate implementation and results directly matches the concept being tested.

**Why B is incorrect:** To avoid using AI tools does not correctly answer the question.

**Why C is incorrect:** To replace certification study entirely does not correctly answer the question.

**Why D is incorrect:** To focus only on hardware does not correctly answer the question.

# Final Answer Key — 3rd Edition

### *Phase 1 — Artificial Intelligence (AI)*

1. A 2. A 3. A 4. A 5. A 6. A 7. A 8. A 9. A 10. A
11. A 12. A 13. A 14. A 15. A 16. A 17. A 18. A 19. A 20. A

### *Phase 2 — Machine Learning (ML)*

21. A 22. A 23. A 24. A 25. A 26. A 27. A 28. A 29. A 30. A
31. A 32. A 33. A 34. A 35. A 36. A 37. A 38. A 39. A 40. A

### *Phase 3 — Deep Learning (DL)*

41. A 42. A 43. A 44. A 45. A 46. A 47. A 48. A 49. A 50. A
51. A 52. A 53. A 54. A 55. A 56. A 57. A 58. A 59. A 60. A

### *Phase 4 — Computer Vision (CV)*

61. A 62. A 63. A 64. A 65. A 66. A 67. A 68. A 69. A 70. A
71. A 72. A 73. A 74. A 75. A 76. A 77. A 78. A 79. A 80. A

### *Phase 5 — Huawei Cloud + DevOps/MLOps*

81. A 82. A 83. A 84. A 85. A 86. A 87. A 88. A 89. A 90. A
91. A 92. A 93. A 94. A 95. A 96. A 97. A 98. A 99. A 100. A